

Introduction 2: transformer $\rightarrow$ IPS ; | LLM.
ICL

Recall the next-token law in LLM. V^T $|V|=K$

Assume the test string $u_{1:T} \sim P_*(u_{1:T}) = \prod_{t=1}^T P_*(u_t | u_{<t})$

Δ_1 good? All test string same law? $\swarrow$ samples $P_*(\cdot | u_{<t}) : V^T \rightarrow [0,1]^K$

Goal: from data $\{u_{1:T}^{(m)}\}_{m=1}^M$ learn $P_\theta(u_{1:T})$ that approximates $P_*(u_{1:T})$

$$\{u_{1:T}^{(m)}\}_{m=1}^M = \prod_{t=1}^{T-1} P_\theta(u_t | u_{<t})$$

Loss fn:
$$\mathcal{L}(P_\theta) = -\mathbb{E}[\log P_\theta(u_{1:T})] = \mathbb{E}_{P_*}[-\sum_{t=1}^T \log P_\theta(u_t | u_{<t})]$$

$$\approx \mathcal{L}_M(P_\theta) = -\frac{1}{M} \sum_{m=1}^M \log P_\theta(u_{1:T}^{(m)}) = -\frac{1}{M} \sum_{m=1}^M \sum_{t=1}^T \log P_\theta(u_t | u_{<t})$$

Estimator: $\hat{P}_{\theta, M} \in \arg\min_{\theta \in \Theta} \mathcal{L}_M(P_\theta)$ $P_\theta(\cdot | u_{<t}) : V^T \rightarrow [0,1]^K$
 Δ_2 How to represent $P_\theta(\cdot | u_{<t})$? Optimization?

$\cdot$ P_* is a minimizer: $\mathcal{L}(P) - \mathcal{L}(P_*) = KL(P_*, P) = \sum_{t=1}^T \mathbb{E} KL(P_*(\cdot | u_{<t}) || P(\cdot | u_{<t}))$

$$KL(P, Q) = \int \log \frac{p}{q} p = \mathbb{E}_p[\log \frac{p(u)}{q(u)}]$$

Δ_3 : Will $\hat{P}_{\theta, M}$ converge as $M \uparrow$? to P_* if Θ changes properly?

Learning theory: classical statistics.

1. In-context Learning:

Assume that there are many next-token laws $P_* \sim \mathcal{P}_0$ unknown

Data $u_{1:T}^{(m)} \leftrightarrow$ sample of $P_*^{(m)}$, (sampled from $\mathcal{U}$, unknown)

Goal: learn the next-token law conditional on $u_{<T}$ $P_\theta(\cdot | u_{<t})$
 $\swarrow$
 context. $\sim P_*(\cdot | u_{<t})$

The same Loss fn and estimator apply

Q: Is $P_*(\cdot | u_{<t})$ a minimizer?

Does $\hat{P}_{\theta, M}$ appr. P_* ? Does the accuracy dep on $\mathcal{P}_0$?

A few simplified Examples.

Example 1. nonparametric regression

$$x \in \mathbb{R}^d, y \in \mathbb{R}^L$$

$$\text{eg. } y = f(x) + \varepsilon$$

Given Data $\{(x_i, y_i)\}_{i=1}^n \sim (X, Y)$

$$\rightarrow (x_{1:n}, y_{1:n})$$

Find a function $y = f(x)$ fitting the data. $f(\cdot | D_n)$

$$\hat{f}_{D_n} = \underset{f \in \mathcal{H}}{\text{argmin}} \quad \frac{1}{n} \sum_{i=1}^n |y_i - f(x_i)|^2 \quad \rightarrow y = \hat{f}_n(x)$$

$$\mathbb{E} \|\hat{f}_{D_n} - f_x\|_{L^2_X}^2 \lesssim n^{-\frac{2s}{2s+d}} \quad f_x(\cdot) = \mathbb{E}[Y|X=\cdot] \in C^s$$

In-context Learning Regression:

Given data $(x_{1:n}, y_{1:n})$ n small; to find $y = f(x | D_n)$

data $\{(x_{1:n}^m, y_{1:n}^m)\}_{m=1}^M \leftarrow M \text{ large} \rightarrow f^{(m)} \sim \mathcal{P}_0$

• We don't use $\hat{f}_{D_n}$ b.c. n too small to est $f_x(x) = \mathbb{E}[Y|X=x]$

• Hope to get posterior of f with prior $\mathcal{P}_0$ & likelihood of D_n

(then, MAP or posterior mean, classification)

• Complexity of $\mathcal{P}_0$: eg. $\mathcal{P}_0: \mathbb{P}(f=f_+) = \mathbb{P}(f=f_-) = \frac{1}{2}$

$$f = \sum_{k=1}^{\infty} w_k \varphi_k$$

$$(\varphi_k(x_i))$$

$$\downarrow x_i \in \mathbb{R}^{d_{\text{new}}}$$

Example 2 Inverse linear regression

Given data $D_n = (x_{1:n}, y_{1:n})$

$$y_i = x_i^T w_x + \varepsilon_i, \quad w \in \mathbb{R}^d, y_i \in \mathbb{R}^L$$

classical:

Estimate w

$$\vec{Y} = \vec{X} w + \vec{\varepsilon}$$

$$\hat{w}_{D_n} = (\vec{X}^T \vec{X}) \mid \vec{X}^T \vec{Y} \xrightarrow{n \rightarrow \infty} w_x$$

$(\vec{X}^T \vec{X} + \lambda I) \mid \vec{X}^T \vec{Y}$ if n not large.

$$\hat{w}_{D_n} = w(x, y); \quad \mathbb{R}^{n \times (d+1)} \rightarrow \mathbb{R}^d$$

ICL:

Given data $D_n = (x_{1:n}, y_{1:n})$ $n \ll d$

data $\bigcup_{m=1}^M D_n^m = (x_{1:n}^m, y_{1:n}^m) \leftarrow w^m \sim \mathcal{P}_0 = \mathcal{N}(0, \Sigma_w)$ — unknown

• will NOT use $\hat{w}_{D_n}$ above, but hope for posterior mean.

$$\hat{w}(x, y) = (x^T x + \frac{1}{\frac{1}{d} \Sigma_w})^{-1} x^T y.$$

unknown, pre-training data

• Complexity of $\mathcal{P}_0$ and condition # of $\mathbb{E} X^T X$?

Example 3: Operator Learning

$A: H \rightarrow H$

Hilbert space H ; linear

classical | Given data $D_n = (u_{i:n}, f_{i:n})$
Est. $\hat{A}_{D_n}$ (then $f = \hat{A}_{D_n} u$)

$A_* u = f + \varepsilon$ noise; u random

$$\text{eg. } A = \sum_{i=1}^{\infty} a_i e_i \otimes e_i$$

$$\mathbb{E} \|\hat{A}_{D_n} - A_*\| \lesssim n^{-\alpha}$$

ICL. | Given $D_n = (u_{i:n}, f_{i:n})$ $\xrightarrow{n \text{ small}}$ find $\hat{A}_{D_n}$
 $\bigcup_n D_n^* = \{(u_{i:n}^a, f_{i:n}^a)\}_{n=1}^M \Leftrightarrow A^{(m)} \sim P_0$